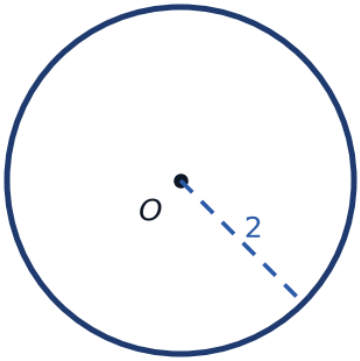
Polar Curves and Derivatives

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

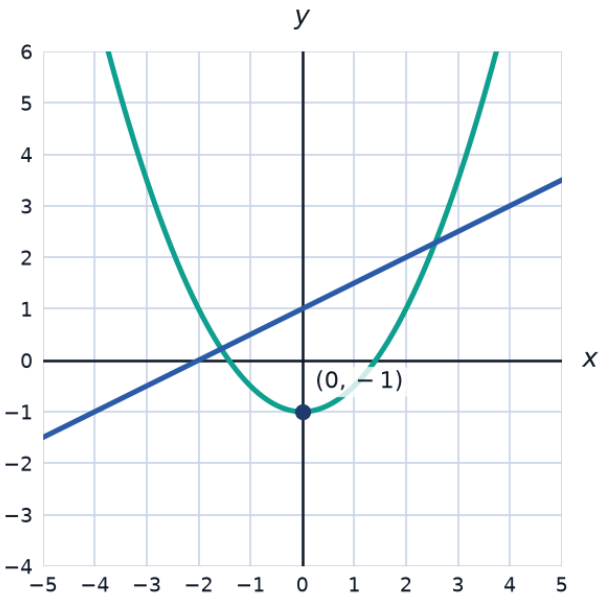
1 **[Foundational]** • **Target: Converting between polar and rectangular coordinates**

Convert the polar equation $r = 4\cos\theta$ to rectangular form, and identify the curve.



2 **[Foundational]** • **Target: Converting between polar and rectangular coordinates**

Convert the polar point $\left(4, \frac{\pi}{3}\right)$ to rectangular coordinates.



3 **[Foundational]** • **Target: Identifying and sketching common polar curves**

Identify the type of curve for $r = 2 + 2\sin\theta$, and state its key feature.

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4 [Foundational] • Target: Identifying and sketching common polar curves

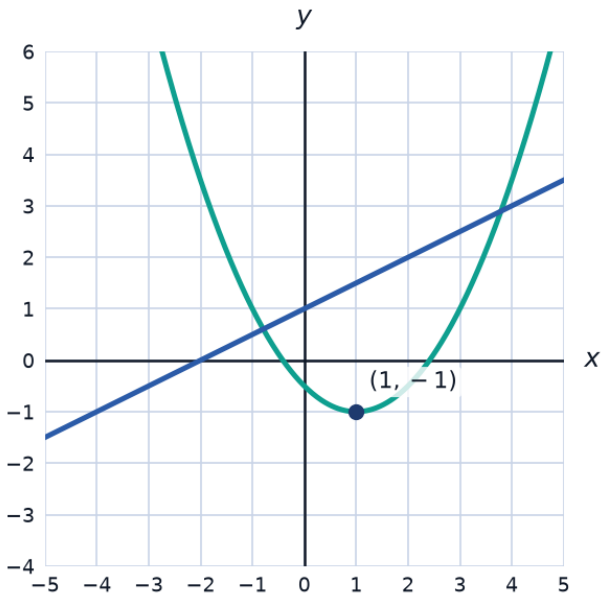
Identify the type of curve for $r = \cos(3\theta)$.

5 [Foundational] • Target: The slope of a polar curve

State the formula for $\frac{dy}{dx}$ of a polar curve $r = f(\theta)$, using $x = r\cos\theta$ and $y = r\sin\theta$.

6 [Foundational] • Target: The slope of a polar curve

Find $\frac{dy}{dx}$ for $r = 2\theta$ at $\theta = \frac{\pi}{2}$.



7 [Foundational] • Target: Finding horizontal and vertical tangents

For the circle $r = 4\cos\theta$, find the value(s) of θ in $[0, \pi]$ where the tangent line is horizontal (where $y = r\sin\theta$ has a critical point).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Symmetry of polar curves

Explain how to test whether a polar curve $r = f(\theta)$ is symmetric about the polar axis (the x-axis).

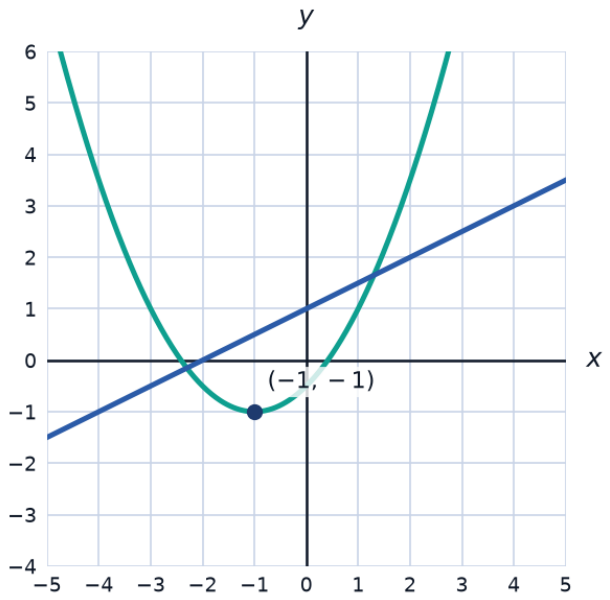
9 [Intermediate] • Target: Symmetry of polar curves

Test whether $r = 3\sin\theta$ is symmetric about the y-axis (replace θ with $\pi - \theta$).

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10 [Intermediate] • Target: More polar-to-rectangular conversions

Convert the rectangular point $(3, 3\sqrt{3})$ to polar coordinates ($r > 0$, $0 \le \theta < 2\pi$).



11 [Intermediate] • Target: More polar-to-rectangular conversions

Convert the rectangular equation $y = x$ to polar form.

12 [Intermediate] • Target: More polar-to-rectangular conversions

Convert the polar equation $r\sin\theta = 4$ to rectangular form, and identify the curve.

13 [Intermediate] • Target: More slope-of-a-polar-curve practice

Find $\frac{dy}{dx}$ for the cardioid $r = 1 + \sin\theta$ at $\theta = 0$.

14 [Intermediate] • Target: More slope-of-a-polar-curve practice

Find $\frac{dy}{dx}$ for the circle $r = 5$ (constant) at any angle θ, and interpret the result using the fact that a circle centered at the origin has radial symmetry.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

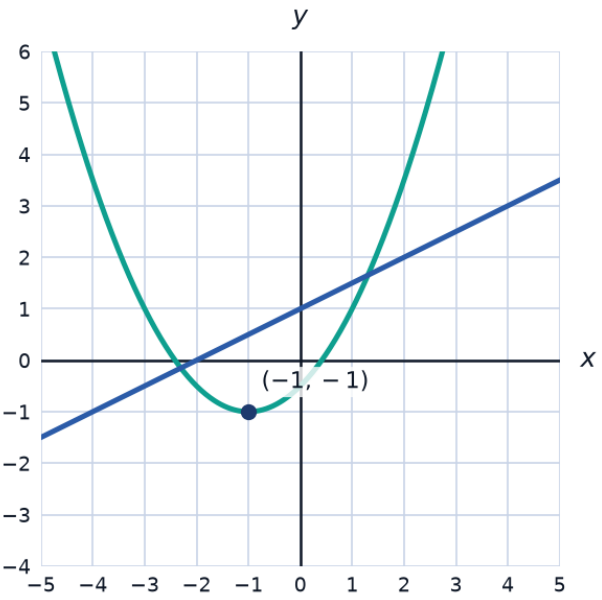
15 [Challenging] • Target: Tangent lines at the pole • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Find the values of θ where the cardioid $r = 1 - \cos\theta$ passes through the pole ($r = 0$), and explain why these give the tangent line directions at the pole.

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16 [Challenging] • Target: Tangent lines at the pole • Cross-Lesson: 1.01 Integration by Parts

Find the values of $\theta \in [0, 2\pi)$ where the rose $r = \cos(2\theta)$ passes through the pole.



17 [Challenging] • Target: Classifying polar curves by equation type • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Classify each polar equation by curve type (circle, cardioid, limaçon with inner loop, rose, or line):

a $r = 5$

b $r = 3 + 3\cos\theta$

c $r = 1 + 3\sin\theta$

d $r = 4\sin(5\theta)$

18 [Challenging] • Target: Number of petals in rose curves • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

State the rule for the number of petals of $r = a\cos(n\theta)$ or $r = a\sin(n\theta)$ depending on whether n is odd or even.

19 [Challenging] • Target: Number of petals in rose curves • Cross-Lesson: 3.01 Parametric Equations

How many petals does $r = 2\sin(4\theta)$ have? How many does $r = 5\cos(7\theta)$ have?

20 [Challenging] • Target: Graphing polar curves point-by-point • Cross-Lesson: 3.02 Vector-Valued Functions

Complete a short table of (θ, r) values for $r = 2 + 2\cos\theta$ at $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$, and use it to describe the cardioid's shape.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Converting between polar and rectangular coordinates

How many petals does the rose $r = \cos(2\theta)$ have?

a. 4

b. 6

c. 7

d. 2

22 [Foundational] • Target: Identifying and sketching common polar curves

How many petals does the rose $r = 4\cos(7\theta)$ have?

a. 14

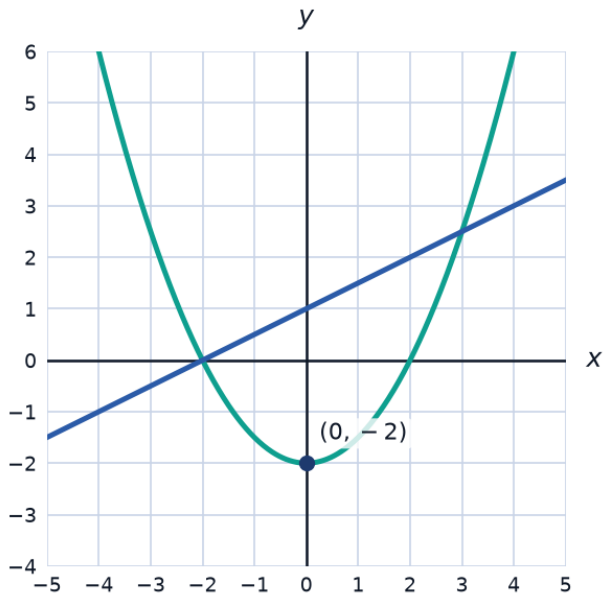
b. 9

c. 7

d. 5

23 [Foundational] • Target: The slope of a polar curve

How many petals does the rose $r = 6\cos(2\theta)$ have?



a. 7

b. 4

c. 2

d. 6

24 [Foundational] • Target: Finding horizontal and vertical tangents

How many petals does the rose $r = 6\sin(7\theta)$ have?

a. 14

b. 5

c. 7

d. 9

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25 [Foundational] • Target: Symmetry of polar curves

How many petals does the rose $r = 2\sin(3\theta)$ have?

a. 6

b. 5

c. 1

d. 3

26 [Foundational] • Target: More polar-to-rectangular conversions

How many petals does the rose $r = 2\cos(4\theta)$ have?

a. 6

b. 4

c. 10

d. 8

27 [Foundational] • Target: More slope-of-a-polar-curve practice

How many petals does the rose $r = 2\sin(6\theta)$ have?

a. 14

b. 12

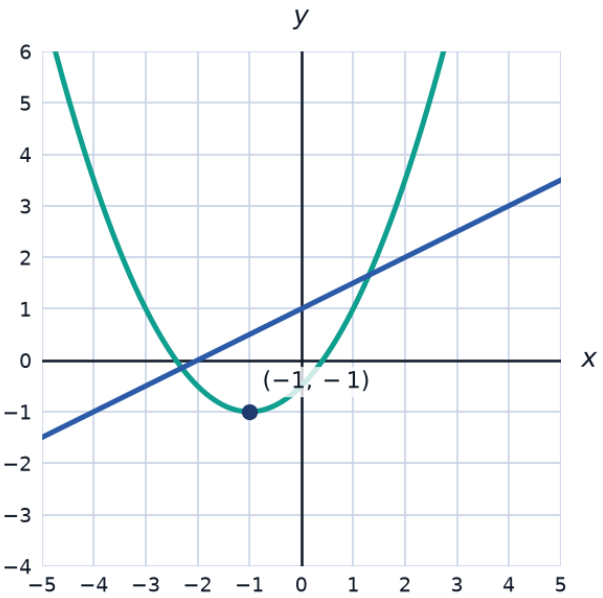
c. 10

d. 6

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28 [Foundational] • Target: Tangent lines at the pole

Convert the polar point $\left(-8, \frac{3\pi}{2}\right)$ to rectangular coordinates.



a. $(-1, 10)$

b. $(8, 0)$

c. $(0, 8)$

d. $(0, -8)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Classifying polar curves by equation type

Convert the polar point $\left(2, \frac{\pi}{2}\right)$ to rectangular coordinates.

a. $(2, 0)$

b. $(0, 2)$

c. $(0, -2)$

d. $(2, 4)$

30 [Intermediate] • Target: Number of petals in rose curves

Convert the polar point $\left(-5, \frac{3\pi}{2}\right)$ to rectangular coordinates.

a. $(5, 0)$

b. $(0, 5)$

c. $(0, -5)$

d. $(2, 3)$

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31 [Intermediate] • Target: Graphing polar curves point-by-point

Convert the polar point $(2, \pi)$ to rectangular coordinates.

a. $(2, 0)$

b. $(-2, 0)$

c. $(-1, 0)$

d. $(0, -2)$

32 [Intermediate] • Target: Concavity considerations for polar curves

Convert the polar point $(-3, 0)$ to rectangular coordinates.

a. $(-3, 0)$

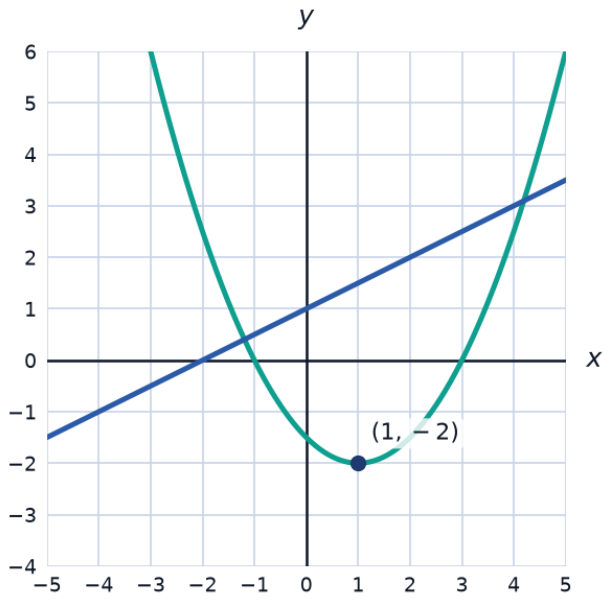
b. $(3, 0)$

c. $(-2, 0)$

d. $(0, -3)$

33 [Intermediate] • Target: Slopes of Tangent Lines to Polar Curves

Convert the polar point $(9, \frac{\pi}{2})$ to rectangular coordinates.



a. $(0, 9)$

b. $(0, -9)$

c. $(9, 0)$

d. $(2, 11)$

34 [Intermediate] • Target: Horizontal and Vertical Tangents on Polar Curves

Convert the polar point $(-5, \frac{\pi}{2})$ to rectangular coordinates.

a. $(-5, 0)$

b. $(0, -5)$

c. $(0, 5)$

d. $(2, -3)$

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35 [Intermediate] • Target: Rate of Change of r and Distance from Origin

Classify the curve $r = 9 + 9\cos\theta$.

a. Circle through the pole

b. Limaçon with an inner loop

c. Limaçon without an inner loop

d. Cardioid

36 [Intermediate] • Target: Arc Length of Polar Curves

Classify the curve $r = 5 + 5\sin\theta$.

a. Limaçon without an inner loop

b. Cardioid

c. Circle through the pole

d. Limaçon with an inner loop

37 [Intermediate] • Target: Converting between polar and rectangular coordinates

Find the smallest positive θ where the rose $r = 4\cos(6\theta)$ passes through the pole.

a. $\frac{7\pi}{12}$

b. $\frac{\pi}{24}$

c. $\frac{\pi}{6}$

d. $\frac{\pi}{12}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Identifying and sketching common polar curves • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Find the smallest positive θ where the rose $r = 9\cos(4\theta)$ passes through the pole.

a. $\frac{\pi}{16}$

b. $\frac{\pi}{4}$

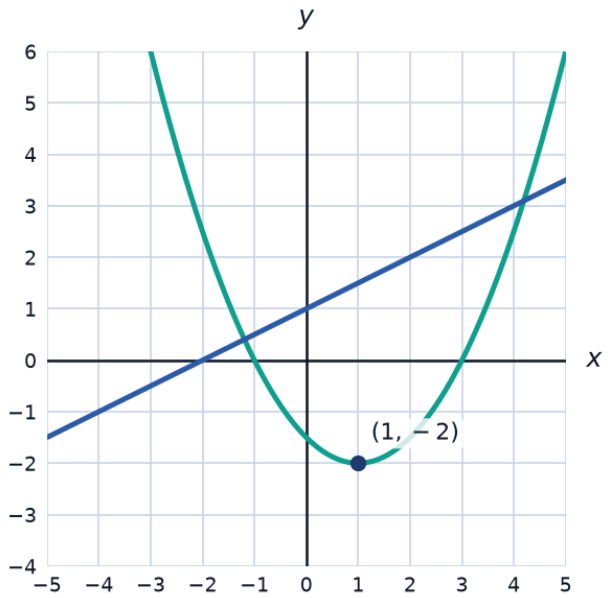
c. $\frac{5\pi}{8}$

d. $\frac{\pi}{8}$

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39 [Challenging] • Target: The slope of a polar curve • Cross-Lesson: 3.01 Parametric Equations

The limaçon $r = 7 + 14\sin\theta$ has an inner loop. Find the smaller positive $\theta \in [0, 2\pi)$ where $r = 0$.



a. $\frac{7\pi}{6}$

b. $\frac{11\pi}{6}$

c. $\frac{5\pi}{6}$

d. $\frac{\pi}{2}$

40 [Challenging] • Target: Finding horizontal and vertical tangents • Cross-Lesson: 3.02 Vector-Valued Functions

The limaçon $r = 8 + 16\sin\theta$ has an inner loop. Find the smaller positive $\theta \in [0, 2\pi)$ where $r = 0$.

a. $\frac{5\pi}{6}$

b. $\frac{\pi}{2}$

c. $\frac{7\pi}{6}$

d. $\frac{11\pi}{6}$

41 [Challenging] • Target: Slopes of Tangent Lines to Polar Curves • Cross-Lesson: 4.02 Area in Polar Coordinates

Which formula gives the slope $\frac{dy}{dx}$ of the tangent line to the polar curve $r = f(\theta)$?

a. $\frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}$

b. $\frac{dr}{d\theta}$

c. $\frac{r\sin\theta}{r\cos\theta}$

d. $\frac{\frac{dr}{d\theta}\cos\theta}{\frac{dr}{d\theta}\sin\theta}$

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42 [Challenging] • Target: Slopes of Tangent Lines to Polar Curves • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

What is the slope of the tangent line to the cardioid $r = 1 + \sin\theta$ at $\theta = 0$?

a. 1

b. -1

c. 0

d. Undefined

43 [Challenging] • Target: Horizontal and Vertical Tangents on Polar Curves • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

At the pole ($r = 0$), what is the slope of the tangent line to the curve $r = f(\theta)$ when $f(\alpha) = 0$?

a. $\tan\alpha$

b. $\cot\alpha$

c. 0

d. Undefined

44 [Challenging] • Target: Rate of Change of r and Distance from Origin • Cross-Lesson: 6.01 Power Series and Interval of Convergence

If $\frac{dr}{d\theta} > 0$ for a polar curve, what does this indicate about the distance from the pole?

a. The distance from the origin is increasing as θ increases

b. The distance is decreasing

c. The curve is concave up

d. The curve is a line

45 [Challenging] • Target: Arc Length of Polar Curves • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Which integral represents the arc length of the polar curve $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$?

a. $\int_{\alpha}^{\beta} \sqrt{r^2 + (\frac{dr}{d\theta})^2}\, d\theta$

b. $\int_{\alpha}^{\beta} \sqrt{1 + (\frac{dr}{d\theta})^2}\, d\theta$

c. $\frac{1}{2}\int_{\alpha}^{\beta} r^2 d\theta$

d. $\int_{\alpha}^{\beta} r d\theta$

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